

Files Transfer and Synchronisation

scp

In case you forget, you can always use scp for transferring files between the client and the server.

```
$ scp <SOURCE> <DESTINATION>
```

If you want to transfer a folder, use -r option like what we normally do for cp.

```
$ scp -r <SOURCE> <DESTINATION>
```

For example, you want to upload a file call model.pth to a folder called my_models under your UPPMAX home directory (So it is ~/my_models/), then you want to move the folder back to the current directory of your local machine, you can do this on your machine:

- To upload, do:

```
$ scp my_model.pth <USERNAME>@pelle.uppmax.uu.se:my_models
```

- To download, do (don't forget the dot in the end, which means current directory):

```
$ scp -r <USERNAME>@pelle.uppmax.uu.se:my_models .
```

rsync

Note

Windows does NOT ship rsync by default, so you need to have either WSL or MinGW or Cygwin installed

rsync is a utility for transferring and synchronising files like scp, but it provides more advanced features like excluding files, reduces latency and so on.

rsync can be used for transferring files like `rsync source target` as scp, but more commonly it is used for synchronising, so the basic operations are more related to a directory like:

- `$ rsync -a source_dir target_dir` for copying a directory, and
- `$ rsync -a source_dir/ target_dir` for all contents in the directory.

Warning

Pay attention to the final slash when using rsync.

- Without the slash, the folder itself will be included for transferring
- With the slash, the folder will not be included. Instead, you will transfer all your files in the folder

If you are not sure about what you typed, use options -anv for a dry-run and verbosed output.

Using the same example above, we have:

- To synchronise all contents from a local folder called my_models/ to server, do:

```
$ rsync -av my_models/ <USERNAME>@pelle.uppmax.uu.se:my_models
```

- To do it in opposite direction but with the directory also:

```
$ rsync -av <USERNAME>@pelle.uppmax.uu.se:my_models .
```